

Rebuilding Rosei's Interband ϵ_2 of Gold in His Own Notation

From Fermi's Golden Rule in k-space — recreating Guerrisi–Rosei–Winsemius Eqs. (1)–(10),
with suggested corrections to three apparent typos: (4), (6), (8)

Ben Shpigel

Department of Physics, Ben-Gurion University of the Negev

July 2026

Introduction

This note rebuilds Rosei's interband $\epsilon_2(\hbar\omega, T)$ of gold from Fermi's Golden Rule, in his own notation, so that the ten numbered equations of Guerrisi, Rosei & Winsemius (“GRW”, *Phys. Rev. B* **12**, 557 (1975)) all come out of one derivation. A few things worth flagging before we start.

Where it goes. Everything feeds two equations, GRW's (7) and (9). Eq. (7) is the thermally weighted, energy-integrated joint density of states,

$$\mathcal{J}_{l \rightarrow u}(\hbar\omega, T) = \int_{E_{\min}}^{E_{\max}} D_{l \rightarrow u}(E, \hbar\omega) [1 - f(E, T)] dE, \quad (\text{GRW 7})$$

and Eq. (9) turns it into the measurable ϵ_2 ,

$$\epsilon_2(\hbar\omega, T) = \frac{8\pi^2 e^2 \hbar^4}{3m^2 (\hbar\omega)^2} \left[|P_X(l \rightarrow u)|^2 g_X(\hbar\omega, T) + |P_L(l \rightarrow u)|^2 g_L(\hbar\omega, T) \right], \quad g_i = N_i \mathcal{J}_i. \quad (\text{GRW 9})$$

The plan is just: build the density of transitions $D_{l \rightarrow u}$, integrate it against occupation to get $\mathcal{J}$ (Eq. 7), and drop that into ϵ_2 (Eq. 9).

X vs. L. In this 1975 paper Rosei really only works out the X point — the saddle and its sub-gap tail, the new physics here. The L point he takes from his earlier papers (Refs. 10 and 12, i.e. Rosei 1974): he says outright that the L procedure “will not be repeated here.” I redo L anyway with the same tools, so both edges sit in one place.

Three equations that look off. Working through the algebra I found that three of the printed equations — (4), (6) and (8) — don't come out as written; most likely they are typos in the 1975 print. I flag each in a red box where it comes up, show the printed form, and suggest the corrected version.

Conventions. Gaussian units (as in the paper); E measured from $E_F = 0$; and — see Sec. 1 — every level and every mass is a positive number, with all signs written out in front.

Contents

1	Notation, and a word on signs	2
2	Fermi's Golden Rule in k-space	2

3 The X point — the part Rosei actually derives	3
3.1 Bands and the constant-energy-difference surface: GRW (1)–(3)	3
3.2 The squared-momentum substitution and the Jacobian	3
3.3 The energy-distributed JDOS: GRW (4)–(5), with a suggested fix to (4)	4
3.4 Topology, the integration limits, and GRW (7)–(8)	6
4 The L point — the part Rosei borrows	7
5 Putting it together: GRW (9)–(10)	9
A Dimensional audit (Gaussian)	10

1 Notation, and a word on signs

E	final-state (upper-band) energy, from $E_F = 0$; $\hbar\omega$: photon energy
$\hbar\omega_{X_6^-}, \hbar\omega_{X_7^+}$	level distances from E_F at X (X_6^- above, X_7^+ below); likewise $\hbar\omega_{L_6^-}, \hbar\omega_{L^+}$
$\mathcal{E}_g^X, \mathcal{E}_g^L$	direct gaps, $\mathcal{E}_g^X = \hbar\omega_{X_6^-} + \hbar\omega_{X_7^+}$, $\mathcal{E}_g^L = \hbar\omega_{L_6^-} + \hbar\omega_{L^+}$
$m_{u\perp}, m_{u\parallel}, m_{l\perp}, m_{l\parallel}$	effective masses (positive; signs written explicitly)
A_i, B_i	$\hbar^2/2m_{i\perp}, \hbar^2/2m_{i\parallel}$ ($i = u, l$): curvature shorthands for derivations
$\Omega_{lu}(\mathbf{k})$	transition energy $\hbar\omega_u - \hbar\omega_l$
$D_{l\rightarrow u}, \mathcal{F}_{l\rightarrow u}, \mathcal{D}_{l\rightarrow u}$	EDJDOS, its mass parameter, and the Jacobian determinant
$\mathcal{J}(\hbar\omega, T)$	thermally weighted EDJDOS integral, GRW Eq. (7)
$ P ^2$	$ \langle u \hat{\mathbf{e}}\cdot\nabla l\rangle ^2$ averaged (length ⁻² ; Sec. 2)
$N_X = 6, N_L = 8$	equivalent half-neighborhoods (GRW's counting)
$f(E, T)$	Fermi–Dirac occupation

Gaussian units, matching the 1975 paper.

Signs and magnitudes. Every level ($\hbar\omega_{X_6^-}, \hbar\omega_{X_7^+}, \hbar\omega_{L_6^-}, \hbar\omega_{L^+}$) and every mass ($m_{u\perp}, m_{u\parallel}, m_{l\perp}, m_{l\parallel}$) here is a *positive number*. Which way a band curves, and which side of E_F a level sits on, is set entirely by the explicit $\pm$ in front of each term — never by a sign hidden in the symbol. (The C&S table I read curvatures off writes some masses with a minus, e.g. $m_{u\parallel} \simeq -0.40$; that only records a downward curvature, and here it becomes a positive $m_{u\parallel} = 0.40$ with the minus moved out in front, as in Eq. (3.1).)

2 Fermi's Golden Rule in k-space

Standard starting point. Minimal coupling in the Coulomb gauge gives $\hat{H}_{\text{int}} = (e/mc) \mathbf{A}\cdot\hat{\mathbf{p}}$ (the A^2 term is interband-inert). Photon momentum is negligible on the zone scale, so transitions are vertical. FGR summed over $\mathbf{k}$ (spin explicit), set equal to the dissipated power $\omega\epsilon_2 E_0^2/8\pi$ per unit volume, gives

$$\epsilon_2(\omega) = \frac{4\pi^2 e^2}{m^2 \omega^2} \frac{2}{(2\pi)^3} \int_{\text{BZ}} d^3k |\hat{\mathbf{e}}\cdot\mathbf{p}_{ul}(\mathbf{k})|^2 \delta(\hbar\omega_u(\mathbf{k}) - \hbar\omega_l(\mathbf{k}) - \hbar\omega) [f(\hbar\omega_l) - f(\hbar\omega_u)]. \quad (2.1)$$

Near a critical point Rosei freezes the matrix element and averages over polarization. With the gradient form $\mathbf{p}_{ul} = -i\hbar\langle u|\nabla|l\rangle$ and

$$|P|^2 \equiv |\langle u|\nabla|l\rangle|^2 \quad (\text{dimension length}^{-2}), \quad \overline{|\hat{\mathbf{e}}\cdot\mathbf{p}_{ul}|^2} = \frac{\hbar^2 |P|^2}{3}, \quad (2.2)$$

this becomes, per critical-point family,

$$\epsilon_2(\hbar\omega, T) = \frac{4\pi^2 e^2 \hbar^2}{3m^2 \omega^2} \frac{2}{(2\pi)^3} N |P|^2 \int_{\text{half-nbhd}} d^3k \delta(\Omega_{lu}(\mathbf{k}) - \hbar\omega) [f(E - \hbar\omega) - f(E)] \Big|_{E=\hbar\omega_u(\mathbf{k})}, \quad (2.3)$$

where N counts equivalent *half*-neighborhoods ($N_X = 6$, $N_L = 8$), which must be paired with the half-space density of states below or a factor of 2 goes missing. The occupation factor is the full $f(E - \hbar\omega) - f(E)$; for the deep d bands it reduces to GRW's $[1 - f]$, but I keep the general form for the non-equilibrium case (f^S , f^P).

3 The X point — the part Rosei actually derives

This is the heart of the 1975 paper, so it gets the full treatment. The X point is a *saddle*, and the payoff of doing it carefully is the sub-gap tail: absorption that starts *below* the nominal gap, purely from geometry, with no broadening invoked.

3.1 Bands and the constant-energy-difference surface: GRW (1)–(3)

GRW's parabolic band pair at X , exactly as they write it (energies from E_F ; $k_{\parallel}$ runs along $\Delta = \Gamma X$, $k_{\perp}$ lies in the zone face). The upper X_6^- (*sp*) band curves *up* in the face and *down* along Δ — a saddle:

$$E = \hbar\omega_u(\mathbf{k}) = \hbar\omega_{X_6^-} + \frac{\hbar^2 k_{\perp}^2}{2m_{u\perp}} - \frac{\hbar^2 k_{\parallel}^2}{2m_{u\parallel}}, \quad [\text{GRW (1)}] \quad (3.1)$$

$$E - \hbar\omega = \hbar\omega_l(\mathbf{k}) = -\hbar\omega_{X_7^+} - \frac{\hbar^2 k_{\perp}^2}{2m_{l\perp}} - \frac{\hbar^2 k_{\parallel}^2}{2m_{l\parallel}}, \quad [\text{GRW (2)}] \quad (3.2)$$

and the lower X_7^+ (*d*) band is a plain maximum (curves down both ways). (Reading off the signs, exactly as promised in Sec. 1: $\hbar\omega_{X_6^-}$ enters +, so X_6^- is above E_F ; $\hbar\omega_{X_7^+}$ enters –, so X_7^+ is below; the one minus in front of the $k_{\parallel}^2$ term of (3.1) is what makes the upper band a saddle rather than a bowl.) Vertical transitions of energy $\hbar\omega$ live on the constant-energy-difference surface (CEDS),

$$\Omega_{lu}(\mathbf{k}) - \hbar\omega = \hbar\omega_u(\mathbf{k}) - \hbar\omega_l(\mathbf{k}) - \hbar\omega = 0. \quad [\text{GRW (3)}] \quad (3.3)$$

Subtract (3.2) from (3.1), with the curvature shorthands $A_i = \hbar^2/2m_{i\perp}$, $B_i = \hbar^2/2m_{i\parallel}$:

$$\Omega_{lu} = \mathcal{E}_g^X + \bar{A}_X k_{\perp}^2 + \bar{B}_X k_{\parallel}^2, \quad \bar{A}_X = A_u + A_l > 0, \quad \bar{B}_X = B_l - B_u. \quad (3.4)$$

The sign of $\bar{B}_X$ decides the whole X story. GRW's own apparatus — the open CEDS, the sub-gap tail, the $-20k_B T$ floor of their Eq. (7), the step-like onset near 1.8 eV — only makes sense if $\boxed{\bar{B}_X < 0}$, i.e. $m_{u\parallel} < m_{l\parallel}$: the *sp* band falls along Δ *faster* than the *d* band (the “flat-*d*” picture). The main text follows this branch; my own C&S extraction currently disagrees on this one assignment, which I flag honestly in Remark 3.1 rather than bury.

3.2 The squared-momentum substitution and the Jacobian

Here is the one trick the whole reduction rests on. Set

$$u = k_{\perp}^2, \quad v = k_{\parallel}^2 \quad (u, v \geq 0).$$

I used to call this a “linearization,” but that name is misleading and I've dropped it: nothing is being approximated. It is an *exact* change of variable. In these squared-momentum variables the two conditions that pin a transition down become honest *straight-line* relations — and, importantly, **each row of the matrix below is simply one of the equations we already have, with the coefficients of $(u, v) = (k_{\perp}^2, k_{\parallel}^2)$ read straight off:**

- the *top* row is the final-state energy, i.e. Eq. (1), (3.1): $E - \hbar\omega_{X_6^-} = A_u k_{\perp}^2 - B_u k_{\parallel}^2 = A_u u - B_u v$;

- the *bottom* row is the transition energy $\hbar\omega$, i.e. the CEDS condition Eq. (3), (3.3), written out with (3.4): $\hbar\omega - \mathcal{E}_g^X = \bar{A}_X k_\perp^2 + \bar{B}_X k_\parallel^2 = \bar{A}_X u + \bar{B}_X v$.

Stacking those two equations is the construction of the transition matrix M , and the Jacobian of the map is nothing but its determinant:

$$\begin{pmatrix} E - \hbar\omega_{X_6^-} \\ \hbar\omega - \mathcal{E}_g^X \end{pmatrix} = \underbrace{\begin{pmatrix} A_u & -B_u \\ \bar{A}_X & \bar{B}_X \end{pmatrix}}_M \begin{pmatrix} u \\ v \end{pmatrix}, \quad \det M = A_u \bar{B}_X + B_u \bar{A}_X = A_u B_l + A_l B_u \equiv \mathcal{D}_X > 0. \quad (3.5)$$

So $\mathcal{D}_X \equiv \det M$ is exactly the Jacobian that will collapse the two δ -functions in (3.9) below; it is built entirely from (3.1) and (3.3). Note it is positive for *either* sign of $\bar{B}_X$ — a small but useful fact. Inverting the 2×2 system,

$$u = \frac{\bar{B}_X (E - \hbar\omega_{X_6^-}) + B_u (\hbar\omega - \mathcal{E}_g^X)}{\mathcal{D}_X}, \quad v = k_\parallel^2 = \frac{A_u (\hbar\omega - \mathcal{E}_g^X) - \bar{A}_X (E - \hbar\omega_{X_6^-})}{\mathcal{D}_X}. \quad (3.6)$$

The second line is the whole content of GRW's Eq. (6). Restoring it to GRW's mass symbols via $\mathcal{D}_X = (\hbar^2/2)^2 \mathcal{F}^{-2}$ (Eq. (3.12) below),

$$k_\parallel^2 = \frac{2\mathcal{F}^2}{\hbar^2} \left[\frac{\hbar\omega - \hbar\omega_{X_7^+} - E}{m_{u\perp}} + \frac{\hbar\omega_{X_6^-} - E}{m_{l\perp}} \right]. \quad [\text{GRW (6), suggested fix}] \quad (3.7)$$

The printed Eq. (6) can't stand as written — almost certainly a typesetting slip.

GRW print this line as

$$k_\parallel = \left(\hbar\omega - \hbar\omega_{X_7^+} + \frac{\hbar^2}{2m_{l\perp}} (\hbar\omega_{X_6^-} - E) - \frac{\hbar^2}{2m_{u\perp}} E \right)^{1/2}. \quad (6, \text{ as printed})$$

You can see the trouble without doing any physics: it sets a wavevector $k_\parallel$ equal to (the square root of) a sum of *energies*. Inside the bracket it also *adds* a bare energy $\hbar\omega - \hbar\omega_{X_7^+}$ to terms of the form $(\hbar^2/2m) \times \text{energy}$, which don't even share its units. The coefficients simply read $\hbar^2/2m$ where they should read $2m/\hbar^2$ for the bracket to come out as $k_\parallel^2 \sim \text{length}^{-2}$. Three small fixes turn the printed line into (3.7): (i) square the left-hand side ($k_\parallel \rightarrow k_\parallel^2$); (ii) invert the coefficients and pull out the overall $2\mathcal{F}^2/\hbar^2$; (iii) restore the missing $1/m_{u\perp}$ on the first term. The three numerator slots then line up with the print term by term, which is how I'm confident (3.7) is what was meant — most likely just a typo, but a disabling one, since the printed formula is unusable as written.

3.3 The energy-distributed JDOS: GRW (4)–(5), with a suggested fix to (4)

Now the central object, and I want to build it rather than drop it on you. Look again at the Golden-Rule expression (2.3): inside the k -integral there is already *one* δ -function — energy conservation, $\delta(\Omega_{lu}(\mathbf{k}) - \hbar\omega)$ — together with the occupation factor $[f(E - \hbar\omega) - f(E)]$, which has to be read at the final-state energy $E = \hbar\omega_u(\mathbf{k})$ and so drifts from point to point across the surface. That final energy is where the temperature gets in (it is the argument of f), so before touching any geometry I want to *sort the transitions by their final energy* E . The clean way to do that is to slip the identity

$$1 = \int dE \delta(\hbar\omega_u(\mathbf{k}) - E) \quad (3.8)$$

into (2.3). This new δ does nothing but *label* each $\mathbf{k}$ by its upper-band energy; it lets me carry the occupation outside the k -integral as a plain function of E ,

$$\int_{k_{\parallel} > 0} d^3k \delta(\Omega_{lu} - \hbar\omega) [f(E - \hbar\omega) - f(E)]|_{E=\hbar\omega_u} = \int dE [f(E - \hbar\omega) - f(E)] \underbrace{\int_{k_{\parallel} > 0} d^3k \delta(\hbar\omega_u - E) \delta(\Omega_{lu} - \hbar\omega)}_{(2\pi)^3 D_{l \rightarrow u}(E, \hbar\omega)}.$$

The purely geometric inner integral — now carrying *both* δ 's: the energy-conservation one that was already there, and the final-energy one I just inserted — is Rosei's central object, the number of $l \rightarrow u$ transition pairs per unit crystal volume, per unit final energy E , and per unit transition energy $\hbar\omega$ (one half-neighborhood, one spin):

$$D_{l \rightarrow u}(E, \hbar\omega) \equiv \frac{1}{(2\pi)^3} \int_{k_{\parallel} > 0} d^3k \delta(\hbar\omega_u(\mathbf{k}) - E) \delta(\Omega_{lu}(\mathbf{k}) - \hbar\omega). \quad (3.9)$$

So the two δ 's aren't pulled from a hat: the second is just the energy-conservation δ already sitting in (2.3), and the first is the resolution of identity (3.8) that bins transitions by their final energy E . That binning is what makes this an *energy-distributed* JDOS, and it is what later lets $\mathcal{J}$ (Eq. 7) weight each energy E by its own occupation $[1 - f(E, T)]$. Let me do this reduction all the way, because it is where both $\mathcal{D}_X$ and Rosei's $\mathcal{F}$ actually come from. Axial symmetry about ΓX gives $d^3k = 2\pi k_{\perp} dk_{\perp} dk_{\parallel}$; in the squared variables $k_{\perp} dk_{\perp} = \frac{1}{2} du$ and $dk_{\parallel} = dv/(2\sqrt{v})$, so the measure becomes

$$\frac{1}{(2\pi)^3} \int_{k_{\parallel} > 0} d^3k = \frac{1}{(2\pi)^3} \frac{\pi}{2} \int_0^{\infty} du \int_0^{\infty} \frac{dv}{\sqrt{v}} = \frac{1}{16\pi^2} \int_0^{\infty} du \int_0^{\infty} \frac{dv}{\sqrt{v}}.$$

The two δ -functions in (3.9) are *exactly the two rows of the transition matrix* (3.5): their arguments are $A_u u - B_u v - (E - \hbar\omega_{X_6^-})$ (top row) and $\bar{A}_X u + \bar{B}_X v - (\hbar\omega - \mathcal{E}_g^X)$ (bottom row). A pair of δ 's whose arguments are linear in (u, v) collapses onto the unique solution (u_*, v_*) of that system — i.e. onto (3.6) — dividing by the modulus of the Jacobian, which is precisely $|\det M| = \mathcal{D}_X$:

$$\delta(A_u u - B_u v - (E - \hbar\omega_{X_6^-})) \delta(\bar{A}_X u + \bar{B}_X v - (\hbar\omega - \mathcal{E}_g^X)) = \frac{1}{\mathcal{D}_X} \delta(u - u_*) \delta(v - v_*).$$

Doing the u, v integrals sets $\sqrt{v_*} = k_{\parallel}(E, \hbar\omega)$ from (3.7), and everything telescopes to

$$D_{l \rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 \mathcal{D}_X} \frac{1}{k_{\parallel}(E, \hbar\omega)}. \quad (3.10)$$

This line is the entire result of the reduction, and notice that it contains only the Jacobian determinant $\mathcal{D}_X$ and $k_{\parallel}$ — there is no $\mathcal{F}$ in it yet. Rosei's $\mathcal{F}_{l \rightarrow u}$ is *not* an independent quantity that I am free to insert; it is nothing but a repackaging of $\mathcal{D}_X$, and here is where it is manufactured. Expand $\mathcal{D}_X$ in the masses using $A_i = \hbar^2/2m_{i\perp}$, $B_i = \hbar^2/2m_{i\parallel}$:

$$\mathcal{D}_X = A_u B_l + A_l B_u = \left(\frac{\hbar^2}{2}\right)^2 \left[\frac{1}{m_{u\perp} m_{l\parallel}} + \frac{1}{m_{l\perp} m_{u\parallel}} \right] = \left(\frac{\hbar^2}{2}\right)^2 \frac{m_{l\perp} m_{u\parallel} + m_{l\parallel} m_{u\perp}}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}}. \quad (3.11)$$

Now *define* Rosei's mass parameter as the pure shorthand $\mathcal{F}_{l \rightarrow u} \equiv \hbar^2/(2\sqrt{\mathcal{D}_X})$. Substituting (3.11), the factor $\hbar^2/2$ cancels top and bottom and the explicit mass form falls right out — this *is* GRW's Eq. (5), now derived rather than assumed:

$$\mathcal{F}_{l \rightarrow u} \equiv \frac{\hbar^2}{2\sqrt{\mathcal{D}_X}} = \left[\frac{m_{l\perp} m_{u\parallel} + m_{l\parallel} m_{u\perp}}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}} \right]^{-1/2}. \quad [\text{GRW (5)}] \quad (3.12)$$

Reading the same definition the other way, $\mathcal{D}_X = \hbar^4/(4\mathcal{F}_{l \rightarrow u}^2)$, and substituting it back into (3.10) rewrites the EDJDOS in Rosei's own notation:

$$D_{l \rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 \mathcal{D}_X} \frac{1}{k_{\parallel}(E, \hbar\omega)} = \frac{\mathcal{F}_{l \rightarrow u}^2}{4\pi^2 \hbar^4} \frac{1}{k_{\parallel}(E, \hbar\omega)} \quad [\text{GRW (4), suggested fix}] \quad (3.13)$$

supported on $u, v \geq 0$, with $k_{\parallel}$ from (3.7). So to be completely clear: $\mathcal{F}$ is never used as an input. The Jacobian hands me $\mathcal{D}_X$; (3.12) is the one-line renaming of it; and $\mathcal{F}$ appears in (3.13) only because I chose to write $\mathcal{D}_X$ that way. Eq. (5) comes out untouched; Eq. (4) does not.

The printed Eq. (4): the prefactor comes out off by a constant — most likely a typo in the paper.

GRW print

$$D_{l \rightarrow u}(E, \hbar\omega) = (8\pi^2 \hbar^2)^{-1} \mathcal{F}_{l \rightarrow u} k_{\parallel}^{-1}. \quad (4, \text{ as printed})$$

This one is subtler than a dropped symbol. If $|P|^2$ is a squared gradient matrix element (or a squared momentum, the other common choice), then the pair {printed (4), printed (9)} doesn't balance dimensionally, whereas (3.13) — which is *forced*, with nothing left to choose, by the definition (3.9) — closes the FGR chain exactly all the way to ϵ_2 (Sec. 5). The two forms differ only by

$$\frac{\text{printed (4)}}{\text{correct (3.13)}} = \frac{2\mathcal{F}_{l \rightarrow u}}{\hbar^2},$$

a *constant for each critical point*. I can't tell from the paper alone whether that is a typesetting slip in the prefactor or just a convention I'm not matching — but it makes no difference to any result: a constant per critical point cannot bend a line shape, so it is absorbed into the fitted strength $S = \mathcal{F}|P|^2$ of GRW's Eq. (10). Every published number keeps its meaning; only the bookkeeping between D , $\mathcal{F}$ and $|P|^2$ shifts. What the derivation above settles is simply that (3.13) is the form consistent with the definition (3.9).

3.4 Topology, the integration limits, and GRW (7)–(8)

Now the part that makes X physically interesting. On GRW's branch $\bar{B}_X < 0$ the CEDS is an *open* hyperboloid for every $\hbar\omega$: transitions exist on *both* sides of $\mathcal{E}_g^X$, and along the CEDS the final energy E decreases without bound as you move away from the top edge ($dE \propto -\mathcal{D}_X dv < 0$). Because nothing bounds the window from below geometrically, it is cut from below by *occupation* alone — GRW use the practical thermal floor $E_{\min} = -20k_B T$, below which $[1 - f] \rightarrow 0$ kills the integrand. The upper edge $E_{\max}(\hbar\omega)$ is geometric, and it changes character at the gap:

The upper edge $E_{\max}(\hbar\omega)$ changes character at the gap. Above it ($\hbar\omega \geq \mathcal{E}_g^X$) the edge is the $k_{\parallel} \rightarrow 0$ circle ($v = 0$ in (3.6)), where $D_{l \rightarrow u}$ diverges like an inverse square root. Below it ($\hbar\omega < \mathcal{E}_g^X$) that circle no longer exists ($v > 0$ everywhere on the CEDS), and the edge is instead the $k_{\perp} = 0$ point ($u = 0$), where $k_{\parallel}$ stays finite and $D_{l \rightarrow u}$ is finite. Writing both cases in one piece,

$$E_{\max}(\hbar\omega) = \hbar\omega_{X_6^-} + \begin{cases} \frac{A_u}{\bar{A}_X} (\hbar\omega - \mathcal{E}_g^X) = \frac{m_{l\perp}}{m_{u\perp} + m_{l\perp}} (\hbar\omega - \mathcal{E}_g^X), & \hbar\omega \geq \mathcal{E}_g^X, \\ -\frac{B_u}{|\bar{B}_X|} (\mathcal{E}_g^X - \hbar\omega) = \frac{m_{l\parallel}}{m_{u\parallel} - m_{l\parallel}} (\mathcal{E}_g^X - \hbar\omega), & \hbar\omega < \mathcal{E}_g^X, \end{cases} \quad [\text{GRW (8), suggested fix}] \quad (3.14)$$

whose lower line (the sub-gap edge) is GRW's Eq. (8).

The printed Eq. (8) has two subscripts interchanged — again, most likely a typo.

GRW print the upper limit as

$$E_{\max} = \hbar\omega_{X_6^-} + (\hbar\omega_{X_7^+} + \hbar\omega_{X_6^-} - \hbar\omega) \frac{m_{u\parallel}}{m_{l\parallel} - m_{u\parallel}} = \hbar\omega_{X_6^-} + (\mathcal{E}_g^X - \hbar\omega) \frac{m_{u\parallel}}{m_{l\parallel} - m_{u\parallel}}. \quad (8, \text{ as printed})$$

The two parallel-mass subscripts are interchanged: the print carries $m_{u\parallel}/(m_{l\parallel} - m_{u\parallel})$ where the clean derivation gives $m_{l\parallel}/(m_{u\parallel} - m_{l\parallel})$ — one consistent $u \leftrightarrow l$ swap. (It’s specific to (8): the subscripts of (6) are *not* swapped, so it isn’t the same slip repeated.) The tell here is physical, not just algebraic. On the $\bar{B}_X < 0$ branch $m_{u\parallel} < m_{l\parallel}$, so the printed denominator $m_{l\parallel} - m_{u\parallel} > 0$ makes $E_{\max}$ climb *above* $\hbar\omega_{X_6^-}$ as $\hbar\omega$ drops below the gap — but the sub-gap CEDS has *no states* up there. The suggested form (3.14) instead puts the sub-gap edge sensibly *below* $\hbar\omega_{X_6^-}$, which is what makes the tail work.

That finite sub-gap edge (3.14) is exactly why the X onset is *step-like* (GRW’s own word) instead of a sharp divergence: at $T = 0$ absorption switches on when the sub-gap edge crosses E_F ,

$$\hbar\omega_{\text{on}} = \mathcal{E}_g^X - \hbar\omega_{X_6^-} \frac{m_{l\parallel} - m_{u\parallel}}{m_{l\parallel}} \simeq 1.94 - 0.17 \times \frac{0.40 - 0.15}{0.40} = 1.83 \text{ eV}, \quad (3.15)$$

which is the piezomodulation threshold, with the famous low-energy tail filling 1.83–1.94 eV purely geometrically (again: no broadening put in by hand).

And finally the thermally weighted integral, verbatim as GRW write it — this is Eq. (7), the first of the two headline results from the opening:

$$\mathcal{J}(\hbar\omega, T) = \int_{E_{\min}}^{E_{\max}} D_{l \rightarrow u}(E, \hbar\omega) [1 - f(E, T)] dE, \quad [\text{GRW (7)}] \quad (3.16)$$

with $D_{l \rightarrow u}$ from (3.13), $E_{\max}$ from (3.14), $E_{\min} = -20k_B T$, and $[1 - f] \rightarrow f(E - \hbar\omega) - f(E)$ once we go out of equilibrium.

Remark 3.1 (The one mass assignment I’m not sure of at X) *Honesty check, because it changes a sign. My extraction of the CES Fig. 5 curvatures assigns $m_{u\parallel} = 0.40$, $m_{l\parallel} = 0.15$ — the opposite $\parallel$ ordering to GRW’s branch — which gives $\bar{B}_X > 0$: a closed window $[\hbar\omega_{X_6^-} - \frac{B_u}{B_X}(\hbar\omega - \mathcal{E}_g^X), E_{\max}]$, no geometric sub-gap tail, and a $\sqrt{\hbar\omega - \mathcal{E}_g^X}$ onset sitting exactly at $\mathcal{E}_g^X$. The catch is that above the gap both branches give the same line shape with the same singular edge (3.14) — the $\parallel$ masses enter only through $\mathcal{F}_X$, which the fit absorbs into S_X — so an optical fit alone cannot tell the two branches apart. Only the sub-gap physics and the extracted $|P_X|^2$ can. Both orderings of my extraction happen to make the d band the lighter one, which is against the flat- d prior, so I suspect a transcription swap in my own mass table is the likeliest fix. Until CES Fig. 5 is re-measured, this note follows GRW ($\bar{B}_X < 0$) and flags every spot where the branch matters.*

4 The L point — the part Rosei borrows

The single most important thing to say about L up front: Rosei does not derive it in the 1975 paper. In his fitting section he states plainly that the L procedure “has been presented in a number of papers and will not be repeated here,” citing his own earlier work (Refs. 10 and 12; the machinery is Rosei 1974, *Phys. Rev. B* **10**, 474). So while X was fresh work, L is inherited wholesale. I re-derive it here anyway — the same squared-momentum substitution gives it in three

lines — so that both edges sit in one framework and you can see *exactly* where L differs from X . And it differs in essentially one sign.

At L the L_6^- conduction band is a *minimum* in both local directions ($k_{\parallel}$ along ΓL) — not a saddle — and the top d band is again a maximum:

$$E = \hbar\omega_u(\mathbf{k}) = \hbar\omega_{L_6^-} + \frac{\hbar^2 k_{\perp}^2}{2m_{u\perp}} + \frac{\hbar^2 k_{\parallel}^2}{2m_{u\parallel}}, \quad (4.1)$$

$$E - \hbar\omega = \hbar\omega_l(\mathbf{k}) = -\hbar\omega_{L^+} - \frac{\hbar^2 k_{\perp}^2}{2m_{l\perp}} - \frac{\hbar^2 k_{\parallel}^2}{2m_{l\parallel}}. \quad (4.2)$$

The only change from X is the sign of the upper-band $k_{\parallel}^2$ term: $+$ here ((4.1)), $-$ at X ((3.1)). That one flip cascades through everything:

$$\Omega_{lu} = \mathcal{E}_g^L + \bar{A}_L k_{\perp}^2 + \bar{B}_L k_{\parallel}^2, \quad \bar{B}_L = B_u + B_l > 0 \quad \text{unconditionally.} \quad (4.3)$$

Because $\bar{B}_L > 0$ *no matter what the masses do*, the CEDS is a *closed ellipsoid*, existing only for $\hbar\omega \geq \mathcal{E}_g^L$: **there is no sub-gap tail at L** — the sharp-versus-tailed contrast between the two edges is baked in right here. The straight-line system is (3.5) with $-B_u \rightarrow +B_u$ and $\bar{B}_X \rightarrow \bar{B}_L$, and the curvature determinant comes out as a *difference* rather than a sum (the ellipsoid’s fingerprint):

$$\mathcal{D}_L = A_u B_l - A_l B_u, \quad \mathcal{F}_L = \frac{\hbar^2}{2\sqrt{|\mathcal{D}_L|}} = \left[\frac{|m_{l\perp} m_{u\parallel} - m_{l\parallel} m_{u\perp}|}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}} \right]^{-1/2}. \quad (4.4)$$

For gold the two products are $m_{l\perp} m_{u\parallel} = 0.70 \times 0.12 = 0.084$ and $m_{l\parallel} m_{u\perp} = 1.03 \times 0.24 = 0.247$, so $\boxed{\mathcal{D}_L < 0}$. Running the inversion (3.6) with the sign of $\mathcal{D}$ reversed (which flips both positivity inequalities) gives

$$k_{\parallel}^2 = \frac{\bar{A}_L}{|\mathcal{D}_L|} (E - E_{\min}), \quad D_{l \rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 |\mathcal{D}_L| k_{\parallel}} = \frac{\mathcal{F}_L^2}{4\pi^2 \hbar^4 k_{\parallel}}, \quad (4.5)$$

$$E_{\min} = \hbar\omega_{L_6^-} + \frac{A_u}{\bar{A}_L} (\hbar\omega - \mathcal{E}_g^L), \quad E_{\max} = \hbar\omega_{L_6^-} + \frac{B_u}{\bar{B}_L} (\hbar\omega - \mathcal{E}_g^L), \quad (4.6)$$

with slopes $A_u/\bar{A}_L = 0.745$ and $B_u/\bar{B}_L = 0.896$ for Au. **Here is the clean punchline of the X vs. L comparison:** the $k_{\parallel} \rightarrow 0$ edge — the inverse-square-root singularity of (4.5) — is now the *lower* limit $E_{\min}$, exactly opposite to X where it was the upper limit. Nothing physical was added; the swap is purely because $\mathcal{D}_L < 0$ while $\mathcal{D}_X > 0$. The L line shape is then

$$\mathcal{J}_L(\hbar\omega, T) = \int_{E_{\min}}^{E_{\max}} D_{l \rightarrow u}(E, \hbar\omega) [1 - f(E, T)] dE, \quad (4.7)$$

and its integrable divergence sits right on the onset edge itself. So: the L edge is *sharp and strong*, the X edge *soft and tailed* — this is GRW’s “splitting of the interband absorption edge,” now with both halves coming out of one formula set instead of two separate papers.

Remark 4.1 (A tension in the L level scheme, recorded on purpose) *The fitted levels ($\mathcal{E}_g^L = 2.45$, $\hbar\omega_{L^+} = 1.56 \Rightarrow \hbar\omega_{L_6^-} = +0.89$ eV) put the whole window (4.6) above E_F , so the Fermi factor is inert at L and the model’s L temperature dependence enters only through broadening and gap shift. But GRW’s own qualitative picture of the L singularity — the onset CEDS sitting tangent to the Fermi surface along the neck — needs L_6^- to be below E_F . Both statements cannot hold in one parabolic model. The fitted scheme is the one the code implements; I record the tension here rather than paper over it.*

5 Putting it together: GRW (9)–(10)

Now we cash in. Insert the revised EDJDOS (3.13) into the FGR result (2.3). The bridge is the definition (3.9) itself, which says $\int d^3k \delta(\Omega - \hbar\omega)[\dots] = (2\pi)^3 \int dE D_{l \rightarrow u}(E, \hbar\omega)[\dots]$ per half-neighborhood; the $(2\pi)^3$ cancels, and

$$\begin{aligned} \epsilon_2(\hbar\omega, T) &= \frac{4\pi^2 e^2 \hbar^2}{3m^2 \omega^2} \cdot 2 \cdot \sum_{i=X,L} N_i |P_i|^2 \mathcal{J}_i(\hbar\omega, T) \\ &= \boxed{\frac{8\pi^2 e^2 \hbar^4}{3m^2 (\hbar\omega)^2} \sum_{i=X,L} N_i |P_i|^2 \mathcal{J}_i(\hbar\omega, T)} \quad [\text{GRW (9)}] \end{aligned} \quad (5.1)$$

— GRW’s Eq. (9), the second headline result, recreated with *no leftover constant*. That clean landing is the payoff, and it happens only if three things are all true at once: (i) $|P|^2$ is the squared ∇ matrix element (2.2); (ii) $D_{l \rightarrow u}$ is the *revised* (3.13), not the printed one; and (iii) spin (the explicit 2) and the half-point counts $N_X = 6$, $N_L = 8$ are kept where shown. (GRW leave the N_i implicit, so their fitted $|P_i|^2$ really means $N_i |P_i|^2$ up to normalization — harmless for the ratio they quote.) The quantities the data actually fix are the strengths,

$$S_X = \mathcal{F}_X |P_X|^2, \quad S_L = \mathcal{F}_L |P_L|^2, \quad [\text{GRW (10)}] \quad (5.2)$$

with $|P_X/P_L|^2 = 0.370$ from their fit to Johnson–Christy. Under the revised (4) the combination the data really pin down is $N_i \mathcal{F}_i^2 |P_i|^2$; it maps onto GRW’s printed S_i through exactly the constant $2\mathcal{F}_i/\hbar^2$ of Remark 5.1, so — to say it one more time — all the published numbers keep their meaning.

Parameters (levels: GRW fit; masses: C&S, with the X row subject to Remark 3.1; $\mathcal{F}$ in units of the free-electron mass):

	$\mathcal{E}_g$ (eV)	upper level (eV)	lower level (eV)	$m_{u\perp}$	$m_{u\parallel}$	$m_{l\perp}$	$m_{l\parallel}$
X (GRW branch)	1.94	$\hbar\omega_{X_6^-} = 0.17$	$\hbar\omega_{X_7^+} = 1.77$	0.19	0.15	0.31	0.40
L	2.45	$\hbar\omega_{L_6^-} = 0.89$	$\hbar\omega_{L^+} = 1.56$	0.24	0.12	0.70	1.03

	$\mathcal{D}$ (units $(\hbar^2/2m)^2$)	$\mathcal{F}/m$	window slopes	singular edge
X	+34.7	0.170	$E_{\max}$ (above gap): 0.38–0.62 [†]	upper ($k_{\parallel} \rightarrow 0$)
L	−7.86	0.357	$E_{\min}$: 0.745, $E_{\max}$: 0.896	lower ($k_{\parallel} \rightarrow 0$)

[†] 0.38 with my $\perp$ assignment ($m_{u\perp} = 0.31$), 0.62 with the fully $u \leftrightarrow l$ swapped one ($m_{u\perp} = 0.19$); $\mathcal{F}_X$ is invariant under the full swap. $\mathcal{D}_X$ and $\mathcal{F}_X$ quoted for the full-swap/GRW assignment; my assignment gives the same values by the symmetry of (3.12).

Scorecard — the ten GRW equations, one by one (the three flagged rows are the ones to watch):

GRW	here	status	issue as printed
(1)	(3.1)	recreated	—
(2)	(3.2)	recreated	—
(3)	(3.3)	recreated	—
(4)	(3.13)	suggested fix	prefactor $(8\pi^2\hbar^2)^{-1}\mathcal{F} \rightarrow (4\pi^2\hbar^4)^{-1}\mathcal{F}^2$; off by $2\mathcal{F}/\hbar^2$ (const.)
(5)	(3.12)	recreated	—
(6)	(3.7)	suggested fix	dimensionally impossible as printed
(7)	(3.16)	recreated	occupation generalized
(8)	(3.14)	suggested fix	$u \parallel \leftrightarrow l \parallel$ subscripts swapped
(9)	(5.1)	recreated	N_i made explicit
(10)	(5.2)	recreated	units note (Rem. 5.1)

Remark 5.1 (The constant that hides the (4) discrepancy) *Collecting the (4) discussion in one place, since it is the subtle one. GRW print $D_{l \rightarrow u} = (8\pi^2\hbar^2)^{-1}\mathcal{F}/k_{\parallel}$; the definition (3.9) forces $D_{l \rightarrow u} = (16\pi^2\mathcal{D})^{-1}/k_{\parallel} = \mathcal{F}^2/(4\pi^2\hbar^4 k_{\parallel})$. The ratio of the two is $2\mathcal{F}/\hbar^2$, a pure constant per critical point. It therefore cannot touch any line shape and is absorbed once and for all into the fitted strength $S = \mathcal{F}|P|^2$ of Eq. (5.2). That is exactly why the discrepancy is so easy to miss and why it costs nothing physically — but the assembly only reproduces GRW’s Eq. (5.1) with no leftover constant if the revised form is used (Sec. 5).*

A Dimensional audit (Gaussian)

Keeping myself honest. $[e^2] = \text{erg cm}$, $[|P|^2] = \text{cm}^{-2}$, $[\mathcal{D}] = \text{erg}^2 \text{cm}^4$, $[\mathcal{F}] = \text{g}$, $[D_{l \rightarrow u}] = \text{erg}^{-2} \text{cm}^{-3}$ (states per volume per energy²), $[\mathcal{J}] = \text{erg}^{-1} \text{cm}^{-3}$.

- (3.13): $\mathcal{F}^2/\hbar^4 k_{\parallel} \sim \text{g}^2/(\text{erg}^4 \text{s}^4 \text{cm}^{-1}) = \text{erg}^{-2} \text{cm}^{-3}$. ✓
- (5.1): $\frac{\text{erg cm} \cdot \text{erg}^4 \text{s}^4}{\text{g}^2 \text{erg}^2} \cdot \text{cm}^{-2} \cdot \text{erg}^{-1} \text{cm}^{-3} = \text{erg}^2 \text{s}^4 \text{g}^{-2} \text{cm}^{-4} = 1$. ✓

References

- M. Guerrisi, R. Rosei, P. Winsemius, Phys. Rev. B **12**, 557 (1975) — the paper recreated here (X derived, L delegated).
- R. Rosei, Phys. Rev. B **10**, 474 (1974) — the L machinery (and the EDJDOS recipe) that GRW borrow.
- N. E. Christensen, B. O. Seraphin, Phys. Rev. B **4**, 3321 (1971) — band structure; source of the masses.
- P. B. Johnson, R. W. Christy, Phys. Rev. B **6**, 4370 (1972) — the data GRW fit.